PX4128 Data Analysis Block B

Background Reading

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Contents

Chapter 5

More on Bayesian Analysis

We are going to take another look at using Bayes theorem – this time not on discrete data points, but for cases where the probabilities are described by a PDF. We will first take a look at some classic (and thus common) forms of PDFs, and then we will discuss how Bayesian Hypothesis Testing works, and how it differs from the classical NHST that we looked at previously.

5.1 Normal likelihood – Normal Prior

- As an introduction to Bayesian analysis with PDFs, we are going to look at a fairly simple (and common) case, in which both the posterior and the likelihood are described by a normal (i.e. Gaussian) distribution.

- Suppose that we have a set of observations $X = x_1, x_2, \ldots, x_n$ of some quantity that we believe to have been drawn from a normal distribution. We can then write,

$$p(x|\theta) = N(\theta, \sigma^2), \tag{5.1}$$

  where θ is the mean value of the distribution (unknown) and σ describes the width (known).

- We can then also write the probability of the mean of the sample, $\hat{X}$, as,

$$p(\hat{X}|\theta) = N(\theta, \sigma^2/n) \tag{5.2}$$

  since as we make more measurements our estimate of the mean gets better (recall the difference between the error and the standard error on the mean!).

- Now suppose that we have some data from a previous study that reports a value for θ of μ_0, with an associated error, σ_0. This study was also subject to random errors, so we are free to write,

$$p(\theta) = N(\mu_0, \sigma_0^2) \tag{5.3}$$

- Bayes theorem allows us to combine this information to determine the PDF of θ. given the data:

$$p(\theta|\hat{X}) = \frac{p(\hat{X}|\theta)p(\theta)}{\int p(\hat{X}|\theta)p(\theta)d\theta} \tag{5.4}$$

where $p(\hat{X}|\theta)$ is the likelihood, $p(\theta)$ is the prior, and the integral on the denominator is the evidence. Since the denominator is the integral over all θ, it is just a constant, and so it doesn't affect the shape of the posterior, only the height. As such, we can ignore it for the moment, and focus on the numerator (i.e. the likelihood times prior).

$$\begin{aligned}
p(\theta|\hat{X}) &\propto p(\hat{X}|\theta)p(\theta) && (5.5)\\
&\propto exp\left[-\frac{(\hat{X}-\theta)^2}{2\sigma^2/n}\right]\ exp\left[-\frac{(\theta-\mu_0)^2}{2\sigma_0^2}\right] && (5.6)\\
&\propto exp\left[-\frac{1}{2}\left(\frac{\hat{X}^2-2\theta\hat{X}+\theta^2}{\sigma^2/n}+\frac{\theta^2-2\theta\mu_0+\mu_0^2}{\sigma_0^2}\right)\right] && (5.7)\\
&\propto exp\left[-\frac{1}{2}\left(\frac{\theta^2}{\sigma^2/n}-\frac{2\theta\hat{X}}{\sigma^2/n}+\frac{\theta^2}{\sigma_0^2}-\frac{2\theta\mu_0}{\sigma_0^2}\right)\right] && (5.8)\\
&\propto exp\left[-\frac{1}{2}\left(\theta^2\left(\frac{1}{\sigma^2/n}+\frac{1}{\sigma_0^2}\right)-2\theta\left(\frac{\hat{X}}{\sigma^2/n}+\frac{\mu_0}{\sigma_0^2}\right)\right)\right] && (5.9)\\
& && (5.10)
\end{aligned}$$

Now, if we consider that θ were a normal distribution, then

$$\begin{aligned}
f(\theta) &\propto exp\left[-\frac{(\theta-\hat{\theta})^2}{2\hat{\sigma}^2}\right] && (5.11)\\
&\propto exp\left[-\frac{1}{2}\left(\frac{\theta^2}{\hat{\sigma}^2}-\frac{2\theta\hat{\theta}}{\hat{\sigma}^2}\right)\right] && (5.12)
\end{aligned}$$

which has the same form as the expansion of the product of the likelihood and prior above. So the posterior must also be a normal distribution, and we can match up the terms of the form θ^2 and 2θ from the relations above. From matching the θ^2 terms, we get,

$$\hat{\sigma}^2 = \left(\frac{1}{\sigma^2/n}+\frac{1}{\sigma_0^2}\right)^{-1}, \tag{5.13}$$

which after a little algebra, gives,

$$\hat{\sigma}^2 = \frac{\sigma_0^2\sigma^2/n}{\sigma_0^2+\sigma^2/n}. \tag{5.14}$$

Similarly, we can match up the 2θ terms to get,

$$\frac{\hat{\theta}}{\hat{\sigma}^2} = \frac{\hat{X}}{\sigma^2/n} + \frac{\mu_0}{\sigma_0^2} \tag{5.15}$$

$$\hat{\theta} = \frac{\sigma_0^2\sigma^2/n}{\sigma_0^2 + \sigma^2/n}\left(\frac{\hat{X}}{\sigma^2/n} + \frac{\mu_0}{\sigma_0^2}\right) \tag{5.16}$$

$$= \frac{\sigma_0^2}{\sigma_0^2 + \sigma^2/n}\hat{X} + \frac{\sigma^2/n}{\sigma_0^2 + \sigma^2/n}\mu_0 \tag{5.17}$$

- The expressions 5.17 and 5.14 give, respectively the **mean** ($\hat{\theta}$) and **variance** ($\hat{\sigma^2}$) of the **posterior** that describes the probability of θ, which in this case, is another normal (Gaussian) distribution. More importantly, we can see that our mean of the posterior, $\hat{\theta}$, depends on the **both** the mean and variance of the data,

$$\hat{X} \;\; \text{and} \;\; \sigma^2 \tag{5.18}$$

and the mean and variance of the prior,

$$\mu_0 \;\; \text{and} \;\; \sigma_0^2. \tag{5.19}$$

This allows to examine the mathematical interplay between the prior and the data (likelihood). Take another look at the expression for the mean of the posterior,

$$\hat{\theta} \;=\; \frac{\sigma_0^2}{\sigma_0^2 + \sigma^2/n}\hat{X} \;+\; \frac{\sigma^2/n}{\sigma_0^2 + \sigma^2/n}\mu_0. \tag{5.20}$$

When the data is good, we see for the 1st term on the RHS, the second term in the denominator is very small, so the fraction in front of $\hat{X}$ tends to 1. At the same time, the fraction in front of the μ_0 term tends to 0. As a result, the estimate of $\hat{\theta}$ from the posterior will favour the estimate from the data, i.e. $\hat{X}$. If, on the hand, the data is sparse, the denominator of the $\hat{X}$ will be large, and the fraction in front of the μ_0 term will tend towards unity. In this case our estimate of $\hat{\theta}$ will favour the prior estimate, μ_0!

- A nice way of thinking about this interplay between the prior and likelihood, is to imagine that the prior is simply adding another data point. If this data point is good (i.e. small variance, such that it is strongly peaked around the mean), then the prior pulls the posterior towards it. If the prior is vague, then it behaves like a data point with a big error, and the posterior relies on the data to provide the underlying shape. This is same behaviour, and indeed very similar maths, to the idea of **weighted averages** that we discussed in Block A.

- At this point, you are probably wondering why we've neglected the evidence! As we mentioned briefly above, the evidence is just a normalisation for the problem. Often, we can ignore the evidence (it really is a terrible name, huh? :). For example when we just cared above about the mean and variance of the posterior, the normalisation is not important. Also, if we simply want to the know the ratio of θ taking two values, we again don't need to know the normalisation of the posterior, so we can ignore the evidence. Similarly, if we are just interested in finding the overall shape of the posterior.

- However, we've established that for the case of a normal likelihood, and a normal prior, the posterior is also normally distributed. Also, we've already worked out the width parameter of the normal, so in this case, it is trivial to normalise the posterior, and thus create a true PDF.

- For problems where we really do want to know the actual probabilities, but where the maths is tricky, we can still work without evaluating the evidence in many cases. For example, one can get a crude approximation to a normalised posterior by first making a **histogram** of the posterior, and then numerically integrating to find the total area under the histogram. If one then divides the original histogram by this area, the result is a **normalised histogram** of the posterior, and as we mentioned early in Block A, this is an approximation to the posterior's underlying PDF.

5.2 Some notes on priors

- A **conjugate** prior is one that has the same functional form as the posterior. In this first example here, we've seen that if the likelihood is normally distributed, then the choice of a normal prior will ensure that the posterior is also a normal, and as such, our prior can be said to be conjugate. This is generally a good thing! Typically, it ensures that the mathematics is possible – it might still not be trivial, but at least it often is analytically tractable. Also, are tricks to dealing with popular families of prior, allowing the analysis to draw on table of standard integrals, etc.

- Even if the true functional form of the posterior is **not** conjugate, it is common practise to approximate the prior distribution with a function that is, simply because it makes the mathematics easier, and it makes the functions behave! Note that the prior is only conjugate when we consider likelihoods of a certain functional form.

- Consider the formula for Bayes Rule again,

$$p(x_i|D) = \frac{p(D|x_i)\, p(x_i)}{\sum_j p(D|x_j)\, p(x_j)}. \tag{5.21}$$

  We have used i to denote that a specific value (or group of values) of x is being tested on the numerator, while the j on the denominator reflects the fact that we need to sum (or integrate, in the case of continuous variables) over **all** possible values of x. Now imagine that we multiply the prior by a constant A, that results in $Ap(x_i) > 1$. We see that the As in the numerators and denominators would cancel! So even though our effective prior has a probability that sums to greater than unity, it doesn't matter, since it would cancel. Such a prior is said to be an **improper prior**, since it is not strictly a probability! Improper priors are used quite a lot, and although they are useful in certain situations, you need to be careful – they do not work work model comparison, for example. If you find yourself resorting to an improper prior, then best to do some background reading on the subject first, to ensure that you use them correctly.

5.3 Beta distributions

- So we have seen that a conjugate prior for a Gaussian/normal likelihood is just another Gaussian/normal. But what about cases where we have a Bernoulli or Binomial distribution as the likelihood? After the normal likelihood, this is probably the second most common form, since it covers a wide range of problems, such the ski test we saw earlier in the course, drug trials, etc. Remember that Bernoulli and Binomial distributions both have the form,
$$p(\nu|N,\theta) \propto \theta^{\nu}(1-\theta)^{(N-\nu)} \tag{5.22}$$
In the case of the Bernoulli distribution, which focuses on a single outcome, the leading constant is 1, while in the case of the Binomial distribution, which accounts for any combination of the ν successes in N trails, the leading constant is given by the binomial factor $\binom{N}{\nu} = N!/\nu!(N-\nu)!$.

- Remember that the functional family that has the same form as 5.22 are called **beta distributions**. The mean and variance of the beta distribution are given by,
$$\hat{\theta}_B = \frac{a}{a+b} \tag{5.23}$$
and
$$\sigma_\sigma^2 = \frac{\hat{\theta}(1-\hat{\theta})}{a+b+1}. \tag{5.24}$$

- One can consider the prior as if it is reporting previously observed data. In this case, we can equate the $a-1$ and $b-1$ in the beta distribution to the ν and $N-\nu$ terms in the Bernoulli and Binomial distributions. This can the be used to guide your choice of a and b. Another way is simply to look at the distributions given in **??** and ask which one represents your belief about the prior best. For example, if you think the coin is a trick/joke coin, but do not know whether it is biased towards heads or tails, then $a = b = 0.5$ is perhaps a good choice! If you think the coin is probably fair, based on a low number (around 5-6) of observations in the past, then $a = b = 4$ might be a good choice, or larger values of a and b if you had a larger sample in the past. Given the number of events in the previous trial, and its outcome, you can use this to calculate your a and b from 5.23 and 5.24

- We can now write out the full form of a Bayesian analysis of a **Bernoulli likelihood with beta prior** as,

$$p(\theta|\nu, N) = p(\nu, N \,|\, \theta)\, p(\theta)/p(\nu, N) \tag{5.25}$$

$$= \frac{\theta^{\nu}\,(1-\theta)^{(N-\nu)}\,\theta^{(a-1)}\,(1-\theta)^{(b-1)}}{B(a,b)\,p(\nu,N)} \tag{5.26}$$

$$= \frac{\theta^{(\nu+a)-1}(1-\theta)^{(N-\nu+b)-1}}{B(\nu+a,\ N-\nu+b)}. \tag{5.27}$$

  Although the powers of θ and $(1-\theta)$ probably made sense there, you are probably wondering where the magic on the denominator came from! The clue is in the way we wrote the powers in the numerator, we you can see that we have deliberately written them in the form a beta distribution, where $a \equiv \nu + a$ and $b \equiv N - \nu + b$, so we can replace the complicated integral in the denominator with the standard normalisation for a beta distribution of the form beta$(\nu+a,\ N-\nu+b)$, which is simply $B(\nu+a,\ N-\nu+b)$.

- Once again, we can look at the interplay between the mean predicted by the data (via the likelihood), and the mean predicted by prior, similar to our analysis for equation 5.17 above. The prior mean is $a/(a+b)$. The mean of the posterior given by substituting (again!) $a \equiv \nu + a$ and $b \equiv N - \nu + b$ into the same equation. Wen can then rearrange the posterior mean to get,

$$\frac{\nu + a}{N+a+b} = \frac{\nu}{N}\,\frac{N}{N+a+b} \;+\; \frac{a}{a+b}\,\frac{a+b}{N+a+b} \tag{5.28}$$

  where we see once again that we weight the data and the prior, by their uncertainty!

5.4 The Bayesian approach to NHST

- Previously we looked at the idea of NHST in the 'classical' framework. As we discussed, there was an issue with this framework in had an **error rate** – the percentage of times that we will incorrectly reject the null hypothesis (i.e. we reject the null hypothesis when in fact it is true). This is often seen as a potential failing of the classical, frequentist, form of hypothesis testing, since you may often reject a model simply due to a single chance (unlikely) measurement, even though that measurement is permitted within the model (it is allowed, just unlikely). In that sense, the observer/experimenter may just be unlucky!

- Another potential problem of the classical NHST is that it does not take the results of any previous experiments into consideration. Sometimes this might be a good thing, particularly if the previous results have been controversial, and you are seeking an independent point of view. However, you can also imagine cases in which you have a great number of pieces of evidence for a particular hypothesis, none of which on their own are particularly convincing but taken together they appear to point towards the hypothesis. For example, suppose each of the pieces of evidence may have only just failed the classical NHST, with their p-values for the null hypothesis scraping in above the standard 5% significance limit. If this happens once, fair enough. But if it happens 5 or 6 times...?

- You may have noticed that the alternative hypothesis H_1 in the NHST did not really enter our analysis – it was an abstract idea, and all the focus was placed on predicting the possibility of the null hypothesis being true. Would it not be preferable in some cases to know the probability of H_1, or perhaps the ratio of the probabilities of H_0 to H_1.

- In many ways, Bayesian inference allows us to get around these problems. First, the outcome of the standard Bayes Rule is a posterior – the probability distribution for the model in question, given the data, **and** how likely you think the model is. With a posterior, different researchers can come along after the experiment and decide which significance level they want to apply. Also, remember that 'Yesterdays's posterior is tomorrow's prior'! The posterior from a previous experiment can be fed as the prior for the next experiment. This flexibility makes Bayesian inference very powerful!

- Here will take another look at a couple of standard procedures in Bayesian inference. We will take a brief look at **estimation approach** (a single prior), and the **model comparison approach** (two priors).

- The estimation approach is perhaps the simplest of the model tests. We are basically asking, given the data is given parameter value credible. The decision rule is therefore:

  A parameter is said to be not credible if it lies **outside** the 95% HDI of the posterior of that parameter. If it lies within the 95% HDI it is said to be credible.

  We can then test whether the value of the parameter predicted by the null hypothesis (let's call it θ_0) lies outside, or within, the 95% HDI. To do this, we can then calculate,

$$p(\theta|x) = \frac{p(x|\theta)p(\theta)}{\int_{\theta_{min}}^{\theta_{max}} p(x|\theta)p(\theta)} d\theta, \tag{5.29}$$

  where x denotes our data. We then examine the posterior to see where our null hypothesis value θ_0 lies. If we have some prior information, then that can be incorporated, to help decide on θ_0. If not (say it is the first time that this experiment is being run, and there is no theoretical reason to favour any particular θ), then we can use a flat prior.

- As an example, consider the case of a series of coin flips. We are given a random coin, and asked to perform a series of 50 flips. We find that heads comes up 35 of the 50 times. Is the coin fair? To to decide whether the coin is fair, we can use 5.29 above. Our likelihood, is given by the Bernoulli distribution,

$$p(\nu = 35 \text{ heads}|N = 50 \text{ flips}, \theta) = \theta^{35} (1-\theta)^{50-35}. \tag{5.30}$$

Since we were given the coin from the sample purely at random, we have no real prior information here, and we are free to chose a flat prior, $p(\theta) = 1$ for $0 < \theta < 1$. To compute the evidence, we need to sum the probabilities of obtaining 35 heads in 50 flips *for all values of* θ – i.e. we sum over the chances of getting the observed series with each of the different coins. Putting this together, we get,

$$p(\theta|N,\nu) = \frac{p(\nu|N,\theta)\, p(\theta)}{\sum_i p(\nu|N,\theta_i)\, p(\theta_i)} \tag{5.31}$$

- The plot of the posterior distribution is given in 5.4, where we have marked the 95% HDI. Here we clearly see that the fair coin, $\theta = 0.5$, falls **outside** the HDI, and so we can reject the null hypothesis that the coin is fair.

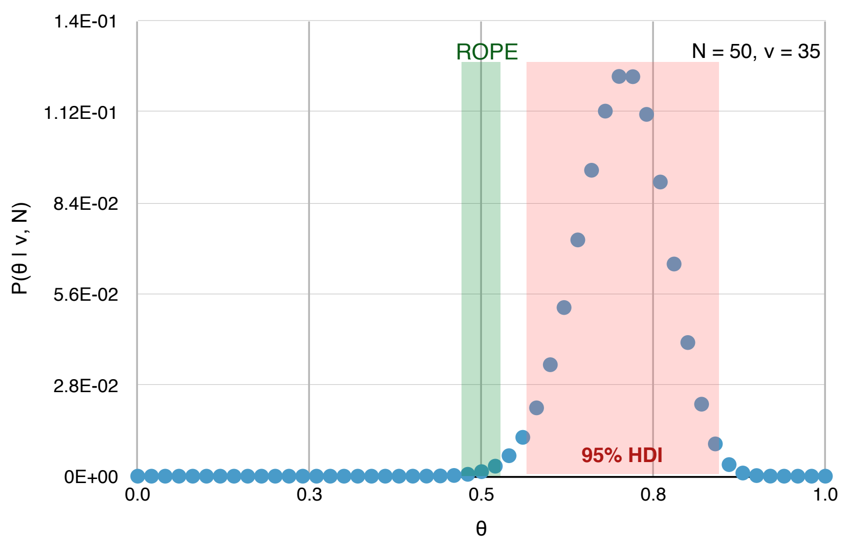

Figure 5.1: The likelihood of the 35 successes in 50 trials.

- With coin tests, and similar types of study, it is clear that we are not really interested in whether θ holds some **exact value**, but rather whether is within a certain range of a desired value. For example, we might consider a 'fair' coin to be one in which the probability of landing heads is $\theta = 0.47$ to 0.53 (i.e. we allow a 0.03 spread in either direction), simply because for our purposes, a coin with p(heads) in this range is good enough. This range of θ that we are willing to accept is called the **Region of Practical Equivalence (ROPE)**. For the values just mentioned, we see from Figure 5.4 that the null hypothesis is **still** outside the HDI, and so we can reject it.

5.5 Differences between Bayesian and Frequentist analysis

At this stage, it is useful to summarise the differences between the **frequentist** and **Bayesian** methods.

1. As we established earlier a frequentist defines probability of an event occurring as the average fraction of times it will occur. The Bayesian stance on probability is much more vague, allowing us to assign probabilities to events that have yet to occur yet, and permits us to express a hunch.
2. The Bayesian analysis explicitly includes a prior. This prior can express the results of previous studies or background knowledge. It can also be used to express a 'belief' about the model under consideration.
3. However probably the most important difference between the two frameworks is how they relate the data and the model:

   The frequentist will ask "how likely is the data given the model"

   The Bayesianist will ask "how likely is model given the data"

   For Bayesians: a probability is a measure of the degree of certainty about values, so the interpretation is that parameters are random and data are fixed. In the Bayes viewpoint PDFs quantify uncertainty in estimating the data and probability $p(x)$ describe how the probability is distributed over possible values of x that might have been measured in a single trial (Figure 5.2(left)).

   For frequentists: a probability is a measure of the the frequency of repeated events, so the interpretation is that parameters are fixed (but unknown), and data are random. Frequentists instead say PDFs quantify variability in a sequence of trials such that $p(x)$ instead describes how the values of x would be distributed among infinte trials N (Figure 5.2(right)).

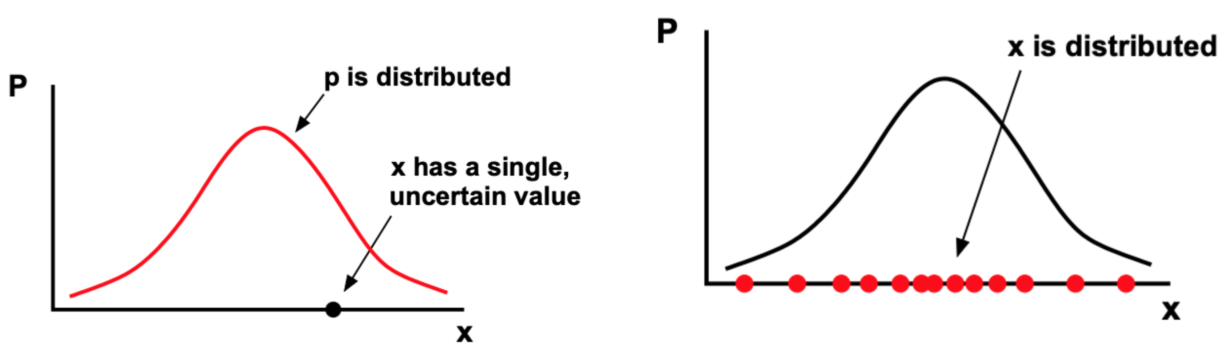

Figure 5.2: How Bayes (left) and Frequentists (right) view PDFs and probabilities.

Chapter 6

Monte Carlo Markov Chains (MCMC)

6.1 A politician stumbles upon MCMC...

We are going to start with a nice example from the Kruschke book (Chapter 7), in which he describes the principles behind MCMC by describing the path that a politician takes on an election campaign

- A politician lives on an chain of 7 islands, than are approximately arranged East-West. An election is coming up in a few years. and so she wants to maximise the amount of time that she spends with the inhabitants of the island chain – i.e. she wants the time she spends on any one island to be proportional to the number of inhabitants on the island. She also needs to be constantly on the move, so that people don't forget about her (or hear to much from her rivals), and so she moves to a new island each day.

- Unfortunately, her retinue are morons, and are unable to keep any information on the populations of the various islands in the chain. All they are able to do is

  1. Ask the mayor of the current island what the population is
  2. Send out to the mayors of the neighbouring islands for their populations.

- The politician is then decides to use this information in the following way:

  1. She first decides whether she is going to move to island to the east (E) or west (W). This is done by simply tossing a (fair) coin: heads = E, and tails = W. If she's on the east-most or west-most island, then the coin toss is still necessary, but she might have to stay were she is if it suggests that she should move to an island that does not exist!
  2. If the population of the proposed island is greater than the population of the current island, then she definitely goes there.

3. However if the population of the proposed island is less than the current island, then she goes there **probabilistically**. To make a decision, she uses a fair spinner with uniformly spaced numbers from 0 to 1 marked on its circumference, which represent a uniform probability distribution p_{spin}. If

$$p_{spin} < \frac{N_{prop}}{N_{curr}}, \tag{6.1}$$

where N_{prop} is the population of the island she's thinking of moving to, and N_{curr} is the population of the island that she is currently on, then she moves, if not, she stays on the current island for another day.

- For the purposes of illustration we will assume that island chain comprises 7 islands, and that the population N of an island is directly proportional to its position in the chain,

$$N_i = i \times 1000 \tag{6.2}$$

Such that the total population of the island chain is $N_{tot} = 28,000$. If things go well, the politician should spend only 1/28 of her time on the first (the west-most) island in the chain, and 5/28 of her time on the 5th island, etc.

- At the start of her campaign, she spends the first day on island 4. The next morning, the coin is flipped to decide whether she heads East or West. The coin lands tails, and so the proposal is to move west to island 3. But the population of this island is less than the population of island 4 – 3000 inhabitants versus 4000, and so she needs to use the spinner to decide if she will move there. The spinner lands 0.3, which is less than 3000/4000, and so she moves to island 3. The next morning the process repeats... etc.

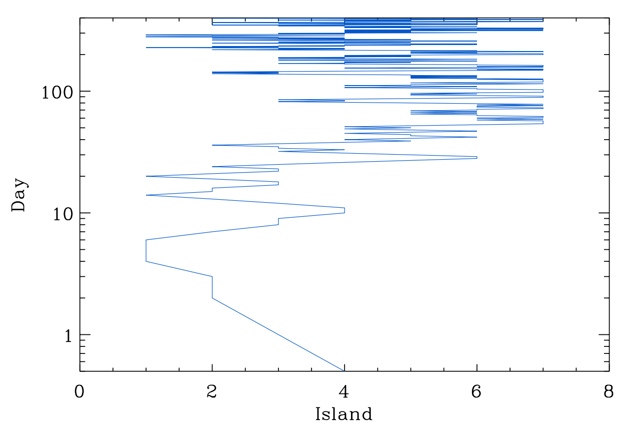

Figure 6.1: The random MCMC walk taken by the politician as she tries to visit each island in proportion to its population.

- We show the path that the politician takes in Figure 6.1. We see that after a brief move to the west, she spends a considerably longer time in the eastern islands (remember that this is also a log axis). Only rarely does she make a visit back to the western islands. Just looking at walk, we get a feeling that she is spending more time in the more heavily populated islands.

- We can quantify this properly by looking at the frequency at which she visits each island. This is shown in Figure 6.2. The blue line shows the visit frequency as recorded after 400 days into the campaign. The black line shows the relative island population as a fraction of the total population of the island chain. This is the politician's **target distribution**. We see that after 400 days the visit frequency is similar in shape to the target distribution – her plan is working! There is however some scatter around the target distribution.

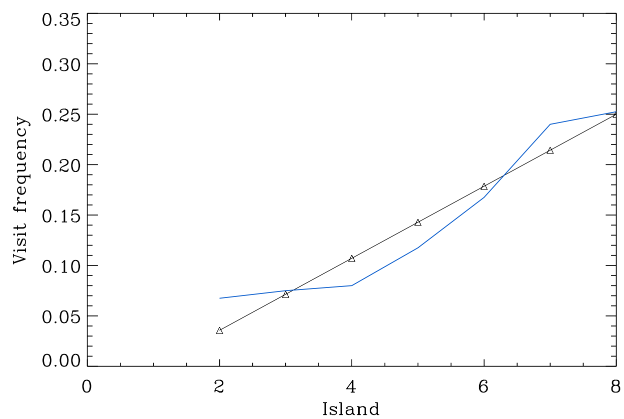

Figure 6.2: The blue line shows the frequency at which the politician visits the islands after 400 days on the move. The black line shows the relative populations of the island chain.

- So what would happen if the politician was to continue this course over many, many days? We show in Figure 6.3 how the walk converges to the target distribution. Even after 4000 days there are significant departures from the the intended frequencies. After 10,000 days they now look indistinguishable.

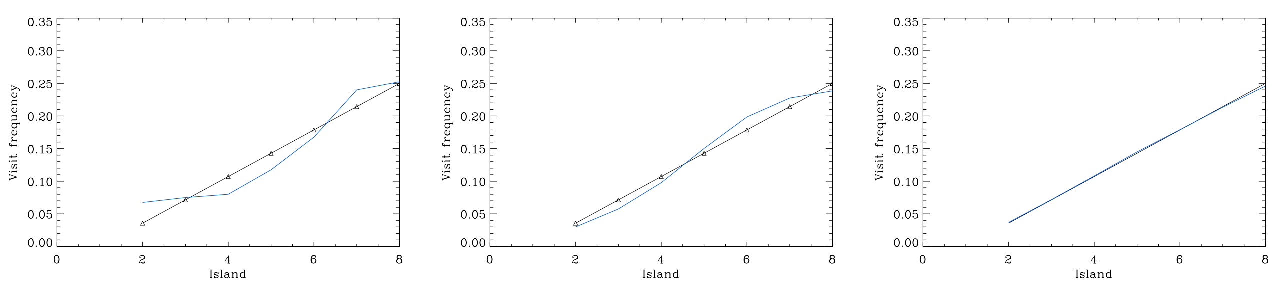

Figure 6.3: We see here how the walk converges to the target distribution over 400 days (left), 4000 days (middle), and 10,000 days (right).

- It is important to remember that the walk shown in Figure 6.1 was just one possible route that the politician could have taken. If any one flip of the coin, or spin of the spinner, had landed differently, then the route would have been different. The walk shown in the above figures was calculated on a computer, with a random number generator playing the role of both the coin and the spinner. In the case of the coin, I simply equated random numbers between 0 and 0.5 with tails (i.e. a proposed move to the west) and 0.5 to 1 with a heads. As you now know from your continual assessment with this course, a random number is initialised with a 'seed' – a number that determines where in the very long sequence of numbers the generator should start. By changing the initial seed (or even changing it at some random point half-way through the calculation) we can generate a different walk. We show the results of this in Figure 6.4. We see that although the walks are different in the details, they still have the same overall features: they spend most of their time on the most populated islands, with only brief forays into the least populated island.

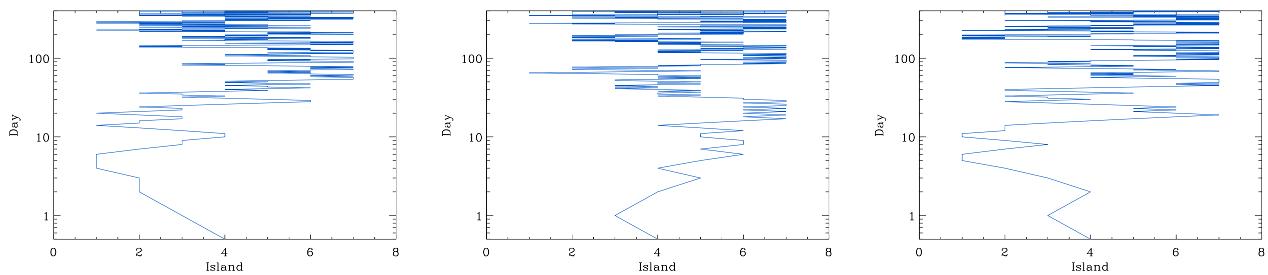

Figure 6.4: Comparison of the walks that can be taken for the island hopping MCMC, for different random initial seeds. The left plot shows our original seed again, and the other two plots show two different seeds.

- So well does the frequencies of these other walks compare to the target distrbution? Was our original attempt in some way a fluke? We show the comparison of the seeds in Figure 6.5. We see that all three seeds are a fair representation of the underlying target distribution. This demonstrates that the politicians decision-making process is stable. In fact, she could also have started on **any** island, and given enough days of campaign, the results would similar to what we see here. However for some starting points the convergence would take longer. We will come back to this later.

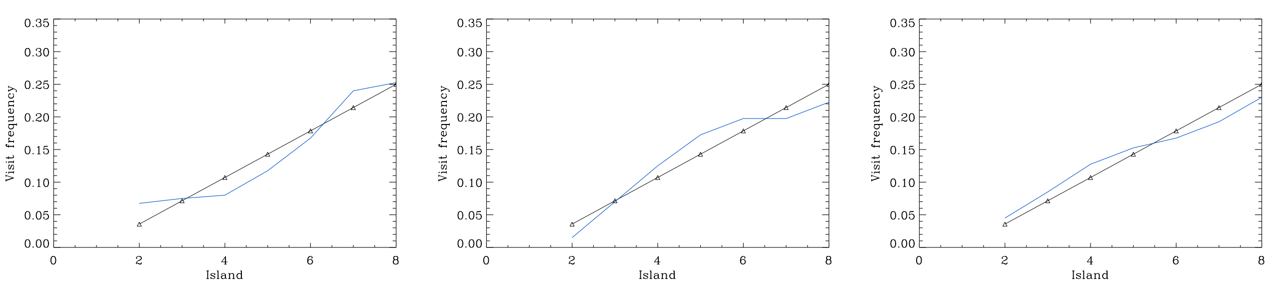

Figure 6.5: Comparison of the frequencies for all three initial random seeds. Again, the original seed is shown on the left.

6.2 A general MCMC algorithm: Metropolis

- The decision-make process used by the politician in the above example is actually known as the **Metropolis algorithm**, and was the first version of the MCMC. It is often cited as one of the top 10 algorithms in computational analysis, and as we will see in the rest of this chapter, it is extremely powerful. It can also be thought of the 'core' MCMC algorithm, since all other forms of MCMC are, in a sense, just a special case of Metropolis.

- The main goal of MCMC is to provide an approximation to a probability distribution. This is extremely useful. Although we have random number generators for uniform, normal, and sometimes binomial/Bernoulli/Poisson distributions, these can only get us so far. For example, the kind of pdfs that commonly appear in Bayesian analysis are often complicated functions, which are hard, or even impossible, to put in an analytic pdf form. MCMC allows us to get around these problems.

- We will now give a general form of the Metropolis algorithm. We are trying to draw numbers from a target distribution that we will denote as $P(\theta)$. The following steps outline the algorithm

  1. Make a first guess at the dependent variable (or variables) $\theta_{current}$.
  2. We then propose to make a random step in the variables to a new location $\theta_{proposed} = \theta_{current} + \Delta_\theta$.
  3. If the value of the function at $\theta_{proposed}$ is greater than that at $\theta_{current}$ – i.e.

     $$P(\theta_{proposed}) > P(\theta_{current}) \tag{6.3}$$

     then we accept a move to the point $\theta_{proposed}$. However, if

     $$P(\theta_{proposed}) < P(\theta_{current}) \tag{6.4}$$

     then we consider the probability of the move to $\theta_{proposed}$ as

     $$p_{move} = \frac{P(\theta_{proposed})}{P(\theta_{current})}. \tag{6.5}$$

     All this can actually be tidied up by simply writing

     $$p_{move} = \min\left(\frac{P(\theta_{proposed})}{P(\theta_{current})}, 1\right) \tag{6.6}$$

  4. If $p_{move} < 1$, then we still have a decision to make. We now use draw a **uniform random number** between 0 and 1 which we will denote as u_{rnd}. If

     $$u_{rnd} \leq p_{move} \tag{6.7}$$

     then the proposed move is accepted. If not, we reject the move and stay where we are.

5. Once the decision is made to accept/reject the point, we update $\theta_{current}$ with the new position (which could be the same position if the move was rejected!) and store its value.
6. Repeat by going back to 1., only now we don't have to guess since we have just moved to $\theta_{current}$.

- One of the most important features about the above algorithm is that the decision to sample a new point $\theta_{proposed}$, depends only on the ratio of the function P at the two locations –**the function need not be normalised**. Indeed, this is the point: we are using MCMC precisely because we could not analytically determine the normalisation factor of P!
- And therein lies the power of MCMC – you can use it to turn any function into a pdf. The values of $\theta_{current}$ that you stored while making the MCMC walk become your random draws from the function P. The only real condition on P is that it is non-negative: we can't have negative probabilities!
- To perform MCMC, we must be able to do several things. We list them here for clarity.
  1. Generate a random value from the proposal distribution (i.e. create $\theta_{proposed}$).
  2. Evaluate the target distribution $P(\theta)$ at any point θ
  3. Draw from a uniform random distribution
- In the list above we used the term **proposal distribution**, which refers to the θ-space from which we are drawing new locations. In most MCMC algorithms, this is done by drawing $\theta_{proposed}$ from a Normal distribution centred on $\theta_{current}$, with some width Δ_θ – i.e. $N_{\theta_{current},\Delta_\theta}(\theta)$. In principle, we are free to chose any function for the proposal distribution, provided that it is **symmetric** around $\theta_{current}$.
- So the Monte Carlo bit makes sense, in that the algorithm depends on random variables. But what about the Markov Chain bit? A Markov Chain (or process) is any series of steps in which each new step has no memory of the step before. Here we see that the Metropolis algorithm is just such a process – the decision to move to $\theta_{proposed}$ depends only on $\theta_{current}$ and $\theta_{proposed}$. After the move, this all forgotten, and a new proposal is made, and so on.

6.3 Quirks and pitfalls

- As you can imagine, it is possible to unknowingly start your MCMC a long way from the underlying peak of the target distribution, and so the first n points in the chain are usually rejected, since their location may be biased. This period of n skipped positions is known as the **burn in**. Given that modern computers are extremely fast, it is common to throw away many thousands of points to ensure that the remaining sample is not biased by the initial conditions. Deciding how many points to skip depends on how well you think the MCMC is doing at sampling your target distribution. We will discuss this more further down. Generally speaking, if you start close to the peak of the target distribution, your burn-in will be shorter than if you start in a region of low (and flat) probability.

- Clearly the value of Δ_θ is also something that needs to considered carefully. If Δ_θ is too small, then the MCMC will explore the parameter space very slowly. It will be accurate, and because $P(\theta_{proposed})/P(\theta_{current}) \sim 1$, the proposals will often be accepted, but clearly it take a very long time to fully sample the distribution. This is a particular problem if you also happen to start your MCMC far away from the peak.

- On the other hand, if you make Δ_θ too large, the ratio of $P(\theta_{proposed})/P(\theta_{current})$ will often be very small, and so there will be very little chance of accepting the proposed move. The result is that you will have a large number of duplicate values in your MCMC sample. One benefit of the large Δ_θ is that it can often move itself out from regions of low probability. However it can also overshoot peaks!

- Another problem associated with the choice of Δ_θ is the clustering of data points. This tends to happen for cases in which the proposal distribution is Normal, and Δ_θ is too small. The cluster arises from the probabilities distribution of the Normal – although it is centred narrowly around $\theta_{current}$ there is a non-zero probability of making a big jump (several σ away). So the MCMC chain tends to loiter in a spot for a long time, before making a big jump to a new patch of the parameter space.

- Another problem with MCMC is that it can get stuck in local maxima. For example, imagine case with a strong prior and conflicting data. In this case there will be (at least) two peaks, one from the data and one from prior. The MCMC walk might loiter around one of the these peaks for a while, before venturing down the slope to lower probability regions of the parameter space. What that occurs, there is a chance that the walk then wander up the other, unexplored peak, but there's also a chance that it simply goes back up the peak that it just came from! Note that this well-explored peak might not be the global maximum – it could be a small, relatively insignificant peak at an overall low probability, but one that we just happened to stumble upon due to our choice of initial starting position. We then have to wait a very long time before the points start to represent the overall distribution.

- So it takes a bit of trial and error to get the standard Metropolis algorithm to converge in the way you want. However, it has the advantage that it is very easy to code, and that computers are now fast enough to allow you explore the parameter space and fine-tune the sampler to hone in on the optimal values. And remember that we often have some idea of the scope and shape of the function that we are trying to sample. However in certain cases, one can take a little of the guess-work out of MCMC, which brings us on to Gibbs sampling...

6.4 Convergence

- During the discussion above, you might be asking yourself a couple of questions: how do I decide when the 'burn in' period is over, and how long does my chain needs to be until it is 'converged'. Indeed, we have been deliberately cagey about mentioning how many N is 'enough', and how to determine the length of the burn in period, etc, for good reason: the mathematics behind these questions is extremely difficult! If you are dealing with complex, multivariate, multi-peaked functions, then you may need to read further than this. Several authors have also made their codes for assessing convergence freely available online. However for simpler case, such you will be examining in this course, it is normally sufficient to apply the techniques listed here.

- The simplest way to assess convergence is to monitor the properties of your sample until any trends have disappeared, such as an increasing mean, or oscillating skewness, etc. It is also good monitor this over different lengths of N within your chain. For example, after 1 million points have been generated, you could try comparing the various moments of the first 1000 points, with the last. This will also start to give you a feel for where the burn-in period ends. Or you plot the moments for each 1000 points, and see at which point in the chain they values start to settle down.

- A more worrying problem is the case that we mentioned above, in which your well sampled chain has become stuck in a not very important peak. Thankfully, a simple solution is at hand: do several chains with different starting positions and/or proposal widths, and allow them to independently explore the pdf. Again you can compare the properties of the various chains at different stages, to see how well they are doing. If one chain has a very different mean, say, than the others, then it is likely that it has become stuck in far-off maximum. Modern CPUs typically have multiple cores, so it is easy to run several chains simultaneously, however you can also simply have you code rotate between chains as it goes, and just wait a little longer.

- For cases where the MCMC results look clumpy, you can perform **thinning**, whereby you only accept each nth value from the chain and throw away the rest. To decide on n, you can either simply look at the data, or perform some sort of correlation statistic on sets of n points, to see where the correlations end (i.e. where the chain starts to jump smoothly). Of course, if you have time, you can also just run the chain again with using a different width for proposal distribution.

- One thing to note is that the values returned from MCMC are not strictly the same as those that you would get from, say, a Gaussian random number generator. This is because successive terms in the chain depend on one another (even when the chain doesn't look clumpy). Thankfully there is another easy way to deal with this. We simply perform our chain, and then randomise the order in which we sample from it.

6.5 Applications of MCMC

- MCMC can be used to generate a random series of numbers from essentially any distribution. For uniform distributions (and simple functions thereof), or normal distributions (an functions thereof) we don't need MCMC – all computing languages come with these functions built in (some will also do Poisson). But for anything else MCMC is extremely useful, since it allows us to turn our simple uniform/normal random number generators into generators for any function.

- One of the most common uses of MCMCs is to obtain a representative sample of points from the posterior,
$$p(\theta|D) \propto p(D|\theta)p(\theta) \tag{6.8}$$
In this case we make $p(D|\theta)p(\theta)$ our target distribution for the MCMC, and the chain of points that results from our converged MCMC run is then a random sample from the Bayesian posterior. This is especially important if our posterior is multivariate (i.e. θ is not one variable, but rather represents a whole host of variables), since then the simpler grid approximations become numerically unwieldy. And once again, we need not worry about trying to normalise $p(D|\theta)p(\theta)$, since MCMC does not care if the function that it is evaluating is normalised or not (only that it is non-negative).

- Once one has a representative sample from the posterior, there are many things that we can do! For example, we can calculate the mean (expectation value) from the points, the variance, etc. That is often enough, which is why we do not always need to produce a normalised posterior.

- Even if a normalised posterior is required, it can often be done simply by creating a normalised histogram of the resulting MCMC points. If the multivariate, this can be done on each variable in turn – that is, first making a histogram of all points that have θ_1, then another with all points that have θ_2, etc. By doing this, you are essentially **marginalising** over the other variables. Once you have your normalised posterior, you can then use it to calculate the HDI of the various marginalised posteriors, which can be useful for hypothesis testing.

- We can also use MCMC to approximate an **integral**. As an example, suppose we want to calculate the mean value of θ – how would be go about that? Intuitively, one would say that we would take the mean of distribution of θ, as yielded by our random draws from the pdf that controls θ or $p(\theta)$ – i.e. we would sum over N draws and divide by N. If N is large enough, and the draws are truly representative of the underlying distribution, then we can write,
$$\int \theta\, p(\theta) d\theta \approx \frac{1}{N} \sum_{\theta_i \sim p(\theta)}^{N} \theta_i \tag{6.9}$$

where we have used $\theta_i \sim p(\theta)$ to stress that the θ_i sample is being drawn from a distribution that is very close to the true pdf of θ. The integral form of the mean should be familiar to you from Block A of this course.

- Expression 6.9 is actually just a special case of a more general equation,

$$\int f(\theta)\, p(\theta) d\theta \approx \frac{1}{N} \sum_{\theta_i \sim p(\theta)}^{N} f(\theta_i) \tag{6.10}$$

in which $f(\theta)$ is simply θ. This more general equation lies at the heart of integration with MCMC, and you should take a while to think about it!

- We can use the above expression to help with **data prediction**. For example, given some data, and an underlying model, we can predict the probability of getting a new data value x from,

$$p(x|D) = \int p(x|\theta) p(\theta|D) d\theta \tag{6.11}$$

This expression might look like it's come from nowhere so we take some time to discuss it. First, note that $p(x|D)$ needs to be a function, somehow, of the model parameters. If there was no data at all, and we simply asked what the probabiiity of predicting a point x is, we would write $p(x) = p(x|\theta)$. This is essentially what we have been doing when we invoked the principle of maximum likelihood in the previous lectures. But the model parameters θ are themselves uncertain, and how well we understand them (i.e. the probability of them being true) depend on how well they fit the data, so we need the $p(\theta|D)$ term to take this into account. Since θ is not a single value, but rather has range of possible values, we need to integrate over these possibilities to work out what the probability of getting x is – hence the integral. Note that by including $p(\theta|D)$, the product of $p(x|\theta)p(\theta|D)$ has units of $1/x\theta$; the integral over θ restores the RHS to the correct units of $1/x$.

- We can now see that the RHS of 6.11 has the same form as 6.10, where $f(\theta) \equiv p(x|D)$ – i.e. the model that we think controls the values – and $p(\theta)$ is now just the posterior for θ **given the data**. In this case, we can then write the probability of observing a data point x as,

$$p(x \mid D) = \int p(x \mid \theta) p(\theta \mid D) d\theta \approx \frac{1}{N} \sum_{\theta_i \sim p(\theta \mid D)}^{N} p(x \mid \theta_i) \tag{6.12}$$

where we see that the draws of θ_i are coming from the posterior for θ. Since MCMC is being used to make the draws, the posterior need not be normalised.

- Another extremely common use of MCMC is to evaluate the evidence, $p(D)$. As we discussed previously, this is often a complex, multidimensional integral. The use of conjugate prior can make the maths a little easier, especially in the case of beta functions. However, in general this term is difficult to evaluate. Recall that the evidence has the form,

$$p(D) = \int p(D \,|\, \theta) p(\theta) d\theta \tag{6.13}$$

Given that this has the same form as 6.10 above, one might simply evaluate the evidence as,

$$p(D) \approx \frac{1}{N} \sum_{\theta_i \sim p(\theta_i)}^{N} p(D \,|\, \theta_i) \tag{6.14}$$

In cases where your prior is fairly flat, or at least broad, this approach should work. However for cases in which the prior is extremely sharply peaked, it can, in practise, take a very long time to converge, since for most of the parameter space $p(D \,|\, \theta_i)$ is very small – only in a small region around the peak in the prior will $p(D \,|\, \theta_i)$ contribute significantly to the sum. Thankfully, there is a trick, that involves sampling from the posterior in clever way! First consider Bayes' rule

$$p(\theta \,|\, D) = \frac{p(D \,|\, \theta)\, p(\theta)}{p(D)} \tag{6.15}$$

We can rearrange this to get,

$$\frac{1}{p(D)} = \frac{p(\theta \,|\, D)}{p(D \,|\, \theta)\, p(\theta)} \tag{6.16}$$

We can now perform the trick introduced by Gelfand & Dey (1994; summarised by Carlin & Louis 2000). We multiply both sides of the equation by 1, but on the RHS, 1 is going to be represented by $\int h(\theta) d\theta$, where $h(\theta)$ is a pdf of some description. We then write,

$$\frac{1}{p(D)} = \frac{p(\theta \,|\, D)}{p(D \,|\, \theta)\, p(\theta)} \tag{6.17}$$

$$= \frac{p(\theta \,|\, D)}{p(D \,|\, \theta)\, p(\theta)} \int h(\theta) d\theta \tag{6.18}$$

$$= \int \frac{h(\theta)}{p(D \,|\, \theta)\, p(\theta)}\; p(\theta \,|\, D)\, d\theta \tag{6.19}$$

$$\approx \frac{1}{N} \sum_{\theta_i \sim p(\theta_i \,|\, D)}^{N} \frac{h(\theta_i)}{p(D \,|\, \theta_i)\, p(\theta_i)} \tag{6.20}$$

So you might be wondering what the point of this was, as this expression now looks significantly more complicated than the previous expression! The answer is that if we chose our pdf $h(\theta)$ to have a shape that is similar to the shape of $p(D \,|\, \theta)\, p(\theta)$ (i.e. the posterior!). By doing this, you effectively help the sum to fully sample the full range of the pdf, rather than just concentrating on the regions around the prior. Note that if you chose $h(\theta)$ to be flat, the expression above would be extremely unstable in the case where $p(D\,|\,\theta)\,p(\theta)$ is strongly peaked, since for many instances the denominator will be very small... You risk having the sum blow up! However, if $h(\theta)$ is chosen carefully, then this is an extremely powerful (and flexible) way of calculating $P(D)$. Note that the above sum gives you the **inverse** of the evidence: i.e. the a multiplicative constant that will normalise your Bayesian analysis.

Chapter 7

Least-Squares Fitting

Here we will review one of the most important aspects of day-to-day data analysis – given data and a model, what the values for the parameters in that model. In other words, **parameter estimation**, one of the most important goals of data analysis.

7.1 Fitting a line to data

- Probably one of the most common things scientists do, is fit a straight line to data, i.e. a linear relation of the form,

$$y = A + Bx. \tag{7.1}$$

  This type of analysis crops up everywhere. Normally what happens is that the experimenters make a scatter plot of x and y, and notice that there's some sort of linear-looking relationship, and decide it would be good to have a mathematical form for this relationship, and so try to fit a straight line. They may even have calculated the covariance matrix from a data set, and realised that two of the variables have a strong correlation (in the above case, variables x and y).

- So given a set of N data points (x_i, y_i), we want to know the values for the intercept A and slope B of the straight line that **best fits the data**. For the moment, we are going to assume that the measurements of y have appreciable uncertainty, while those of x do not. Further, we are going to assume that the errors associated with y are **normally distributed**. A sketch of such a situation is shown in Figure 7.1, where the error bars show the standard deviation. Although this might seem like an unlikely scenario, this situation is often the case – remember that errors combine in quadrature and so one error tends to dominate over the other. We are also going to assume at the moment, that the errors all have the same magnitude. Later we will relax these constraints.

- Once again, we are going to look towards the principle of **maximum likelihood** for help. If we knew what A and B were, we could, for a given value of x_i, compute the

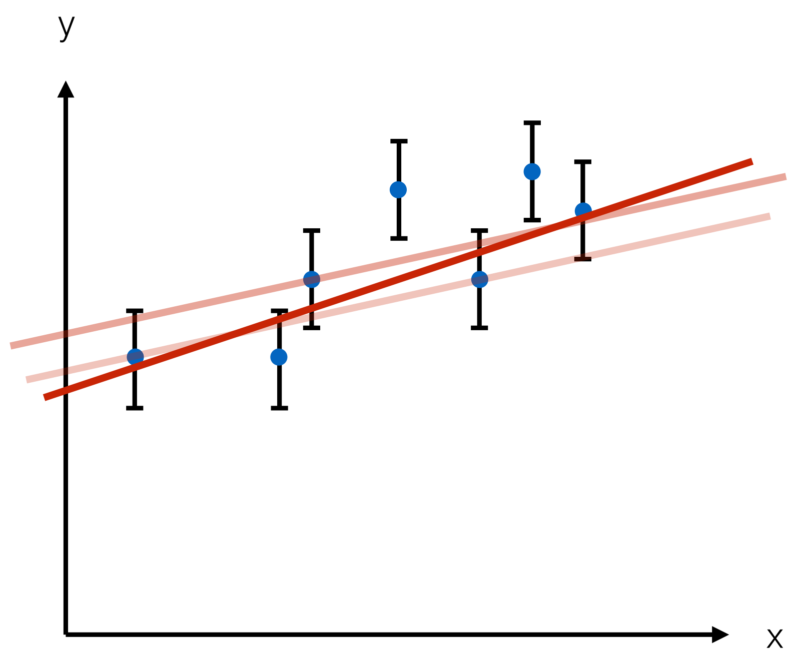

Figure 7.1:

corresponding value of y_i,

$$\text{true value for } y_i = A + Bx_i. \tag{7.2}$$

The measured values of y_i that we see in Figure 7.1, are therefore drawn from a normal distribution of width σ_y that is centred on the true value y_i. So we can write the probability of obtaining any given measurement y_i as,

$$p_{A,B}(y_i) \propto \frac{1}{\sigma_y} e^{-(y_i - A - Bx_i)^2/2\sigma_y^2}. \tag{7.3}$$

The subscripts A and B indicate that this probability depends on the unknown values of A and B.

- Taking this a stage further, we can now write the probability of obtaining our entire set of measurements as

$$p_{A,B}(y_1, y_2, \ldots, y_N) = p_{A,B}(y_1) \cdots p_{A,B}(y_N) \tag{7.4}$$

$$\propto \frac{1}{\sigma_y^N} e^{-\chi^2/2} \tag{7.5}$$

where χ^2 is our old friend,

$$\chi^2 = \sum_{i=1}^{N} \frac{(y_i - A - Bx_i)^2}{\sigma_y^2}. \tag{7.6}$$

- As before, we are interested in recovering the values of A and B that maximise the probability of obtaining the set of the observed measurements, (x_i, y_i). The joint probability of the measurements given in Equation 7.4 is obtained when χ^2 is a minimum, so we need to differentiate Equation 7.6 with respect to our unknowns A and B, and set the differentials equal to zero, as we did before:

$$\frac{\partial\chi^2}{\partial A} = \frac{-2}{\sigma_y^2}\sum_{i=1}^{N}(y_i - A - Bx_i) = 0 \tag{7.7}$$

$$\frac{\partial\chi^2}{\partial B} = \frac{-2}{\sigma_y^2}\sum_{i=1}^{N}x_i(y_i - A - Bx_i) = 0 \tag{7.8}$$

This results in a pair of simultaneous equations for A and B,

$$AN + B\sum x_i = \sum y_i, \tag{7.9}$$

and

$$A\sum x_i + B\sum x_i^2 = \sum x_i y_i, \tag{7.10}$$

where we have dropped the limits on the sumation for sake of clarity.

- These equations can then be solved for A and B to get,

$$A = \frac{\sum x^2 \sum y - \sum x \sum xy}{N\sum x^2 - (\sum x)^2} \tag{7.11}$$

$$B = \frac{N\sum xy - \sum x \sum y}{N\sum x^2 - (\sum x)^2} \tag{7.12}$$

Together, these equations allow us to calculate the best fit to the line $y = A + Bx$. The resulting line called the **least squares fit**, or **line of regression** of y on x. Now that we have A and B, we naturally want to estimate the uncertainties in these values. But first we need to discuss the uncertainty in σ_y...

7.2 Uncertainties in y

- In the analysis above, we assumed that the values of y_i were distributed around the true values with spread of σ_y. You will notice however, that our estimates of A and B do not actually depend on this number (or numbers, in the case where σ_y is different for each point). However the true values of y_i are predicted **by** the line which A and B describe, and so the deviations of our measured values y_i should be depend on A and B. This immediately suggests that a good way to estimate σ_y is from,

$$\sigma_y = \sqrt{\frac{1}{N}\sum_{i=1}^{N}(y_i - A - Bx_i)^2} \tag{7.13}$$

which is the usual root-mean-square deviation from the mean, where in this case, the mean is defined by the best estimate of the line we have just fitted (i.e. the values A and B). One could check this: as usual, the best guess is the one that maximises the probability of obtaining the data, and so one could differentiate Equation 7.4 with respect to σ_y and set to zero. The result would be the relation for σ_y above.

- However, Equation 7.13 has a problem, in that it violates the **degrees of freedom**. The problem is that both A and B have been defined from the data already, and so we need to replace the $1/N$ with $1/N-2$. to get,

$$\sigma_y = \sqrt{\frac{1}{N-2}\sum_{i=1}^{N}(y_i - A - Bx_i)^2}. \tag{7.14}$$

This makes sense: if you have only two data points, then the best guess for the line will have A and B such that the line goes **exactly** through both points – there is no scatter! As soon as we have 3 data points, then at least 1 will not sit on the line, and the idea of σ_y becomes meaningful.

7.3 Uncertainties in A and B

- Now that we know the uncertainties in our measurements y_i, it is possible to calculate the uncertainties in A and B. To do this, note that both A and B are well-defined functions of y_i. This means that we can use the standard error propagation formula that we encountered in Block A. Using this, we get,

$$\sigma_A = \sigma_y\sqrt{\frac{\sum x^2}{N\sum x^2 - (\sum x)^2}} \tag{7.15}$$

and

$$\sigma_B = \sigma_y\sqrt{\frac{N}{N\sum x^2 - (\sum x)^2}} \tag{7.16}$$

- In the above derivations, we have so far assumed that there were errors in y, but no errors in x. What if it is the other around? The answer is that we simply exchange x for y in our analysis, and proceed as usual.

- What if there are uncertainties on **both** x and y? The general situation of performing a least-squares fit to a curve where there are errors on both variables is quite complicated, and even controversial. However for the special case of straight line, such as we have presented here, the uncertainties in both x and y make very little difference. To see this imagine that there is no error in y, but x has an error δx. This point lies off the line, at a distance δy – it produces an equivalent error in y. For the case of a straight line, the relationship is,

$$\delta y(\text{equiv}) = \frac{dy}{dx}\delta x. \tag{7.17}$$

The standard deviation σ_x is the root-mean-square value of δx that would result from repeating this measurement for each of our points, and so,

$$\sigma_y(\text{equiv}) = \frac{dy}{dx}\sigma_x. \tag{7.18}$$

This is true for any function that has no second derivative – remember the idea of 'bias' from earlier – but in the special case of a straight line, $^{dy}/_{dx}$ is simply our fit parameter, B. Thus the problem of fitting a line with uncertainties in x but none in y is the same as the problem of fitting a line with uncertainties in y but none in x.

- In the case where there are uncertainties in both x and y, we simply translate the σ_x to σ_y. Accounting for the intrinsic errors in y, this yields,

$$\sigma_y(\text{equiv}) = \sqrt{\sigma_y^2 + (B\sigma_x)^2} \tag{7.19}$$

where we have combined the errors in quadrature in the usual way. Clearly, if we are not interested in fitting a straight line, then this expression may be more complicated, since we would need to retain the $^{dy}/_{dx}$, rather than using the B.

- What about the case in which the errors on y_i are not constant? In this case we need to use the idea of **weighted least squares**, that incorporates the weighted averages that we encountered in Block A. By carrying the weights $w_i = 1/\sigma_i^2$ (these are on y_i) through the analysis, one ends up with the following expressions,

$$A = \frac{\sum wx^2 \sum wy - \sum wx \sum wxy}{\sum w \sum wx^2 - (\sum wx)^2} \tag{7.20}$$

and

$$B = \frac{\sum w \sum wxy - \sum wx \sum wy}{\sum w \sum wx^2 - (\sum wx)^2}, \tag{7.21}$$

with associated uncertainties,

$$\sigma_A = \sqrt{\frac{\sum wx^2}{\sum w \sum wx^2 - (\sum wx)^2}} \tag{7.22}$$

and

$$\sigma_B = \sqrt{\frac{\sum w}{\sum w \sum wx^2 - (\sum wx)^2}} \tag{7.23}$$

Basically we see that every term gets a w and N is replaced by $\sum w$. This replacement takes the place of the leading σ_y term that we had in the case of the homoscedastic errors.

7.4 Least squares fits to other curves

7.4.1 Polynomials

- Now that we have covered the basic ideas of linear regression with the easy case of a straight line, we can generalise the method to more complex functions. The first, most simple generalisation is moving from a straight line, to a polynomial (since a straight line is a first order polynomial!), which we will express as,

$$y = A + Bx + Cx^2 + \cdots + \alpha_n x^n. \tag{7.24}$$

  A classic example that we are all familiar with is the height y of falling body in the absence of air resistance,

$$y = y_0 + v_0 t - {}^1\!/_2 g t^2 \tag{7.25}$$

- For the purposes of discussion here, we will focus on the second-order case of a quadratic, but it is straightforward to extend the analysis to more higher orders. For the quadratic, we have,

$$y = A + Bx + Cx^2 \tag{7.26}$$

  we will again assume that in our set of N data points (x_i, y_i), the y_i values are all equally uncertain, and the x_i values are all exact. Also, we will again assume that the uncertainties in the y_i values are normally distributed, where the central (true) value is governed by Equation 7.26 and thus determined by A, B and C. These assumptions once again allow us to express the probability of obtaining the data points y_i as,

$$p(y_1, y_2, \ldots, y_N) \propto e^{-\chi^2/2} \tag{7.27}$$

  where χ^2 has now become,

$$\chi^2 = \sum_{i=1}^{N} \frac{(y_i - A - Bx_i - Cx_i^2)^2}{\sigma_y^2} \tag{7.28}$$

  Once again, we want to maximise the probability by minimising χ^2, so we need to differentiate with respect to our unknowns A, B and C, and set the derivatives equal to zero. This time, we obtain 3 equations:

$$AN + B\sum x + C\sum x^2 = \sum y \tag{7.29}$$

$$A\sum x + B\sum x^2 + C\sum x^3 = \sum xy \tag{7.30}$$

$$A\sum x^2 + B\sum x^3 + C\sum x^4 = \sum x^2 y. \tag{7.31}$$

  These simultaneous equations for A, B and C are referred to as the **normal equations**. Clearly for higher order cases, and especially those where some constants are 0, it is easier to package the whole thing up in matrix notation, and solve the system of equations using linear algebra and an off-the-shelf matrix inversion program!

- In principle, a similar method can be applied to **any** function $y = f(x)$, however in practise it is often difficult, or even impossible, to solve the resulting set of normal equations. We will see that it is possible to get approximate solutions using numerical methods.

- However, provided $f(x)$ depends **linearly** on the unknown parameters A, B, C, etc. then it is, in principle, **always** analytically possible to solve the resulting normal equations. Actually, this class of problems is very large. For example the function,

$$y = A \sin x + B \cos x \tag{7.32}$$

  is linear on both A and B. The reason being that both $\partial y/\partial A$ and $\partial y/\partial B$ are linear functions of A and B, and so a unique solution is possible.

- This is a problem: one of the most important functional forms in physics is non-linear: the exponential form,

$$y = Ae^{Bx}. \tag{7.33}$$

  For example, the attenuation of the intensity I in a medium (such as a gas) falls off with distance x from the source with,

$$I = I_0 e^{-\mu x} \tag{7.34}$$

  and the charge Q in a capacitor drains with time as,

$$Q = Q_0 e^{-\lambda t}. \tag{7.35}$$

  So what do we do?

- If the equation can be **linearised**, then we can proceed as normal. The easiest way to linearise a exponential function is to take the natural logarithm of both sides:

$$z = \ln y = \ln A + Bx \tag{7.36}$$

  We can now work in (x_i, z_i) space, instead of (x_i, y_i) space, and proceed as normal.

- However, we need to be careful! Our previous derivations for A and B above have assumed that all the values of $y_1, y_2, \ldots, y_N$ we all equally uncertain (i.e. **iid**!). But clearly this is not the case for our new linearlised system, even though it could have been for the original data – our errors in the original data, σ_y, have been transformed,

$$\sigma_z^2 = \left(\frac{dz}{dy}\right)^2 \sigma_y^2 = \frac{\sigma_y}{y}. \tag{7.37}$$

- The answer is use the weighted least-squares that we defined above! We can then work out the σ_z for each point z_i, which in this case is going to be a function of both y_i and σ_y

- **Note:** technically, our errors are still a function of y, so if the errors are large, then there could be considerable **bias**. If that's the case, then the above expression will only approximate the error, and we need to include the bias explicitly.

- Also note that we could have had a function with more than one variable – a **multivariate** function, such as,
$$z = A + Bx + Cy. \tag{7.38}$$
In this case, we would proceed directly as before! Since the equations are linear in the fit variables (A, B, C), this is still a standard case, which results in a set of 3 normal equations that can be solved analytically.

7.5 Maximum likelihood

- We have seen from the above discussion of least-squares fitting how the χ parameter represents the residuals between the data, and the fit. We focused on χ-squared minimisation, because it maximised the likelihood. We looked at the special case of linear regression, but the maximum likelihood idea is, in fact much more powerful, as we will discuss here.

- In the event of no prior knowledge – i.e. a **flat prior** – The likelihood of a set of model parameters α given the data X, is simply,
$$L(\alpha) = p(X|\alpha) = p(x_1|\alpha) \times p(x_2|\alpha) \times \cdots \times p(x_N|\alpha) \tag{7.39}$$
$$= \prod_{i=1}^{N} p(x_i|\alpha), \tag{7.40}$$
assuming the data points were again independent on one another.

Example: Prosecutor's Fallacy

A local newspaper reports that a "John Doe" drowned. The probability that someone ζ will die in the US due to cause H can be written as $P(\zeta|H)$. According for the Centers for Disease Control and Prevention (CDC), the number of deaths in the US due to accidental drowning in 2011 is 3,556. That year the population was approximately 3.12×10^8. What is the likelihood that John Doe's death was due to accidental drowning?

It is tempting to do the following:

$$P(\zeta|H) = \frac{3556}{3.12 \times 10^8} = 1.14 \times 10^{-5}$$

$$\mathcal{L}(x|H) = \mathcal{L}(\text{John Doe}|\text{accidental drowning})$$

$$\mathcal{L}(x|H) = P(\text{John Doe}|\text{accidental drowning}) = 1.14 \times 10^{-5}$$

Ie a prosecutor would conclude that the likelihood of drowning accidentally is very low, implying foul play.

We can resolve this by using likelihood ratios instead.

The FBI tell us that in 2011, there were 15 homicides by drowning. The probability then that a person ζ will die by murder is

$$P(\zeta|H) = P(\zeta|\text{homicidal drowning}) = 4.8 \times 10^{-8}$$

The likelihood for John Doe now has two cases:

$$\mathcal{L}x|H) = \begin{cases} \mathcal{L}(\text{John Doe}|\text{accidental drowning}) = 1.14 \times 10^{-5} \\ \mathcal{L}(\text{John Doe}|\text{homicidal drowning}) = 4.8 \times 10^{-8} \end{cases}$$

The ratio of the two give the likelihood ratio (LR) $LR = 240$. Therefore John Doe is 240 times more likely to have drowned due to an accident compared to a homicide.

7.5.1 Maximum Likelihood for χ^2

Going back to χ^2, for a Gaussian error distribution, the likehood function is written as:

$$p(x_i|\alpha) = \frac{1}{\sqrt{2\pi}\sigma_i} exp\left(-\frac{(x_i - \mu_i(\alpha))^2}{2\sigma_i^2}\right) \tag{7.41}$$

for the probability of each point, so we can write the likelihood function as,

$$L(\alpha) = e^{-\chi^2/2} \times \prod_{i=1}^{N} \frac{1}{\sigma_i} \times (2\pi)^{-N/2}. \tag{7.42}$$

We can now take the natural log of this to get,

$$-2\ln L = \chi^2 + 2\sum_{i=1}^{N} \ln \sigma_i + N\ln(2\pi) \tag{7.43}$$

To maximise $L(\alpha)$, we then need to minimise $\chi^2 + 2\sum_{i=1}^{N} \ln \sigma_i$. The maximum likelihood α_{ML} is then the one that satisfies,

$$\frac{\partial}{\partial \alpha}\left[-2 \ln L(\alpha)\right] = 0 \tag{7.44}$$

and the variance of α_{ML} this given by,

$$\mathrm{var}(\alpha_{ML}) \approx \frac{2}{\frac{\partial^2}{\partial\alpha^2}\left[-2\ln L(\alpha)\right]_{\alpha=\alpha_{ML}}} \tag{7.45}$$

- You might be wondering what the point in this is, since it seems to just be another way of writing the χ^2 test, but in a more obscure and general way! Well, there are actually several reasons for this.

  1. The first is that the basic idea of maximum likelihood is not only true for normal distributions, but can apply to any distribution of the errors, so in the case where the errors are not normally distributed, it can come to your aid. The expressions above for α_{ML} and its variance were derived for a normal distribution about the mean, so they only apply where that is a valid assumption.
  2. You are free to add prior information if you have some to hand!
  3. You will notice in the case of the normal distribution derived above that $L(\alpha)$ depends on the errors in the points σ_i. In the case were the errors are not functions of the fit parameters α, then we recover what we had before. However if the fit parameters affect the data (i.e. $\sigma_i = f(\alpha)$), then the expressions above will take that into account. If the dominant source of error is still from the data, and the model only adds to the error, then you can generally just use the χ^2 minimisation (provided, of course, that the data errors result in measurement values that are normally distributed around the mean). If the errors are unknown, and depend strongly on the model, then maximum likelihood is a better option.

- Poisson statistics is a good example of a case where the fit parameters can affect the error bars. This is especially true in cases where the data have low mean count rates (i.e. low signal to noise).

Chapter 8

GENERAL LEAST SQUARES AND BEYOND

8.1 Do we have a good fit?

We can use the goodness of fit test χ^2 to determine if we have a good fit to the data based on our line. **Need to first introduce the idea that one can simply use small departures in best fitting parameters** A **and** B **(dA and dB) to explore the chi-square landscape!**

- The discussion of least-squares fitting was based around the idea that we want to maximise the likelihood that a given model was responsible for the data. In the special case of a normal distribution, we seen that this reduced minimising the χ^2, which can be thought of as the residuals between the data points and the model.

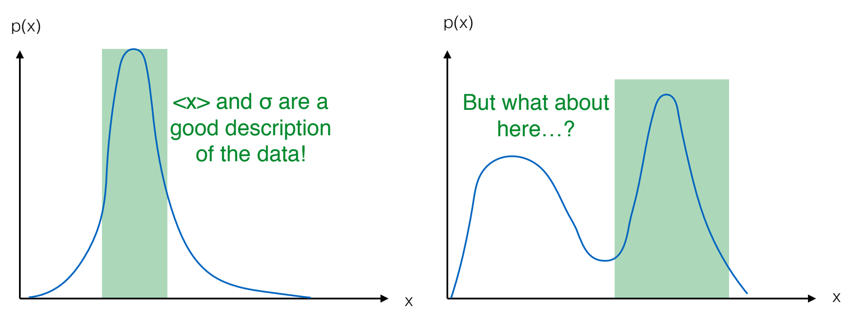

Figure 8.1:

- However we need to be careful of how to interpret the 'best fit' to the data. For example, have a look at the two PDFs of some parameter x in Figure 8.1. In the left hand panel, we show an example of a strongly peaked PDF. In this case, the mean and variance make sense, and we could argue that they are good description of the data (even though there is some skew to the right of the mean). However in the right-hand panel, we see a very different case. Although there is still a well-defined peak – and thus a clear maximum likelihood — one could argue that the associated peak is not really representative of the underlying distribution. The other obvious problem is that there is a second peak! Although this peak is lower, it is clearly much broader. The problem that the data analyst then faces, is how to deal with such a distribution! It might be better **not** to invoke maximum likelihood in this case, and simply randomly sample from the PDF if the data is required elsewhere.

- A similar situation can occur in the χ^2-space. Imagine that instead of simply solving the normal equations that result from the χ^2 minimisation for the straight line fit problem, one instead simply varies the values of A and B systematically, and computes the corresponding values of χ^2 (one could do this the $L(\alpha)$ too!). We can then make a 3D plot of the χ^2 values, such as that shown in Figure 8.2. Here, the contours represent lines of constant χ^2, with light blue showing low values of χ^2 and darker-blue through to black showing progressively higher values.

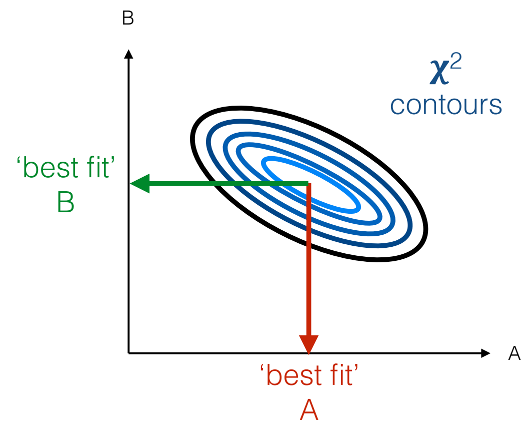

Figure 8.2:

- In this particular case, we have drawn a situation in which there is only 1 peak, and the peak is relatively well defined. However there could have several peaks! Or, as we show in Figure 8.2, the central contour of χ^2 is quite elongated, suggesting that perhaps while B is well-constrained, the values of A may be less so! Plotting the χ^2 surface is extremely useful, as it allows you to check how well the model is fitting the data.

- The other common use of the χ^2 surface is to 'read off' the errors in the fit parameters. For example, let us suppose we have a series of fit parameters α and we have data in which we know the errors (or intrinsic spread) to be normally distributed. We can then either maximise $L(\alpha)$ or minimise χ^2 to find α. The $1-\sigma$ confidence interval on α includes the central 68% of the area under $L(\alpha)$. However, this is also equivalent to the range of α covered by $\Delta\chi^2 = 1$, as we show in Figure 8.3. In the case were we have only 1 parameter in α, then the $k-\sigma$ interval is given by the range of α enclosed by $\Delta\chi^2 = k^2$, i.e.,

$$\Delta\chi^2 = 1 \quad \text{for } 1-\sigma, \ 68\% \text{ probability} \tag{8.1}$$
$$\Delta\chi^2 = 4 \quad \text{for } 2-\sigma, \ 95.4 \ \% \text{ probability} \tag{8.2}$$
$$\Delta\chi^2 = 9 \quad \text{for } 3-\sigma, \ 99.73\% \text{ probability} \tag{8.3}$$

So if the maths to compute σ for α turns out to be tricky, one can simply create a plot of $\chi(\alpha)^2$, and read off the appropriate values for the standard deviation.

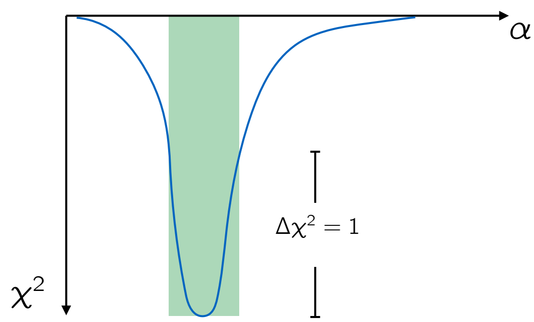

Figure 8.3:

- When α contains more than 1 parameter, as was the case for the straight-line fit, the result is a little more complicated, since now we are integrating probability in 2 dimensions. In practise, this is a little involved (See Press et al. for details), but will can summarise the result here in the following table for you to use.

Table 8.1: $\Delta\chi^2$ thresholds for M parameter $k-\sigma$ confidence regions

$k-\sigma$	probability	$M=1$	2	3	4
$1-\sigma$	68%	1	2.30	3.53	4.72
$2-\sigma$	95.4%	4	6.17	8.02	9.70
$3-\sigma$	99.73%	9	11.8	14.2	16.3

- For 2 a parameter fit, the region enclosed by constant χ^2 is an area. For a 3 parameter fit, it is a volume, etc.

8.2 Orthogonal parameter fitting

- In the case of the 2 parameter fit of a straight line, we looked at how exploring the χ^2 space can tell us about how well the parameters have been estimated. Another feature of the χ^2 space is that it can show us how the model parameters **correlate** with one another.

- Take the straight line fit once more, as shown in Figure 7.1. We see that as increase the slope B of the line, the intercept A will decrease. As we reduce the slope of the line, the intercept will rise. Thus, for the case of the straight line, we say that the model parameters A and B are **correlated**. Any error we make in estimating A will be knocked on to B, which in some sense will have to compensate. These model parameters are therefore not really independent, and so if we were to forward model new data sets, we would need to be careful when selecting from the probable values of A and B.

- The correlation in the parameters affects the orientation of the of ellipses in the χ^2 surface. In the case of the straight line fit, the parameters are inversely correlated, and so the χ^2 ellipses point down, as shown in Figure 8.2. If an increase in one parameter were to cause an increase in the other, then the χ^2 ellipses would point up.

- For cases in which the χ^2 contours are vertical or horizontal, then we can say the fit parameters are **orthogonal** – i.e. they are independent of one another. Such a setup is shown in Figure 8.4. For all values of A, all the χ^2 profiles in B have the same minimum (i.e. they point to the same B). Likewise, for a all values of B, all the χ^2 slices in A have a minimum at the same value of A.

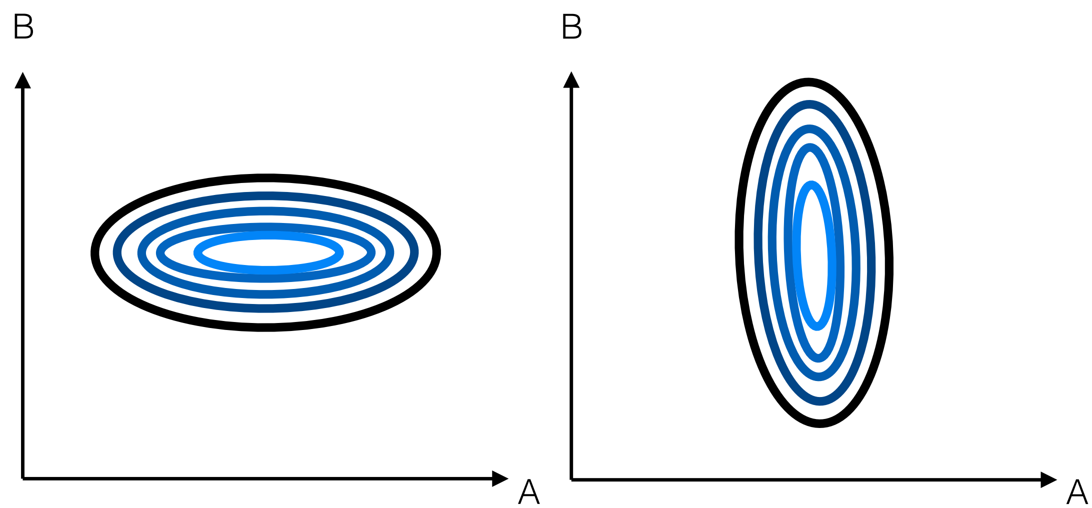

Figure 8.4: Example of χ-squared space if parameters are independent of one another - we would say the fit parameters are orthogonal.

- For the simple case of a straight line, it is actually fairly trivial to orthogonalise the model parameters. Instead of fitting the line $y = A + Bx$, we can instead fit $y = A + B(x - \hat{x})$, where $\hat{x}$ is just the mean of the x measurements, calculated in the standard way. If we look at the diagram in Figure 8.5. We now see that the parameters A and B are independent: the value of A simply controls the height at which the line passes the point $x = \hat{x}$, and the value of B controls the angle (slope) of the line as it passes through.

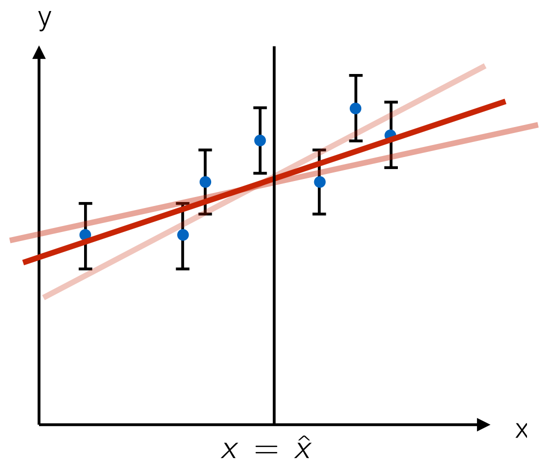

Figure 8.5:

8.3 General least-squares fitting and the Hessian matrix

- Let us go back to our (weighted) straight line fit for $y = A + Bx$. The χ^2 that we want to minimise is,

$$\chi^2 = \sum_{i=1}^{N} \left(\frac{y_i - (A + Bx_i)}{\sigma_i} \right)^2 \tag{8.4}$$

To make things a little easier, we will drop the subscripts on the variables and errors. We then minimise χ^2 with respect to the model parameters in the usual way,

$$0 = \frac{\partial \chi^2}{\partial A} = -2 \sum (y - A - Bx)/\sigma^2 \tag{8.5}$$

and

$$0 = \frac{\partial \chi^2}{\partial B} = -2 \sum x(y - A - Bx)/\sigma^2 \tag{8.6}$$

which then results in the normal equations,

$$A \sum 1/\sigma^2 + B \sum x/\sigma^2 = \sum y/\sigma^2 \tag{8.7}$$

$$A \sum x/\sigma^2 + B \sum x^2/\sigma^2 = \sum xy/\sigma^2 \tag{8.8}$$

As you will remember your first year maths (!), you can package up these equations into matrix notation, which in this case would give,

$$\begin{pmatrix} \sum 1/\sigma^2 & \sum x/\sigma^2 \\ \sum x/\sigma^2 & \sum x^2/\sigma^2 \end{pmatrix} \begin{pmatrix} A \\ B \end{pmatrix} = \begin{pmatrix} \sum y/\sigma^2 \\ \sum xy/\sigma^2 \end{pmatrix} \tag{8.9}$$

This equation then has the form,

$$\underline{H}\,\underline{\alpha} = \underline{C}(y) \tag{8.10}$$

where we have stressed the $\underline{C}$ is the matrix that holds the the y terms by writing $\underline{C}(y)$. The solution is simply,

$$\underline{\alpha} = \underline{H}^{-1}\,\underline{C}(y) \tag{8.11}$$

where $\underline{H}$ is the **Hessian** matrix and $\underline{H}^{-1}$ is its inverse, which is given (in this simple case) by,

$$\underline{H}^{-1} = \frac{1}{\Delta}\begin{pmatrix} \sum x^2/\sigma^2 & -\sum x/\sigma^2 \\ -\sum x/\sigma^2 & \sum 1/\sigma^2 \end{pmatrix} \tag{8.12}$$

where the determinant $\Delta = \sum x^2/\sigma^2 \sum 1/\sigma^2 - (\sum x/\sigma^2)^2$. We then can write the solution as,

$$\begin{pmatrix} A \\ B \end{pmatrix} = \frac{1}{\Delta}\begin{pmatrix} \sum x^2/\sigma^2 \sum y/\sigma^2 - \sum x/\sigma^2 \sum xy/\sigma^2 \\ -\sum x/\sigma^2 \sum y/\sigma^2 + \sum 1/\sigma^2 \sum xy/\sigma^2 \end{pmatrix} \tag{8.13}$$

which, you will see are exactly those given by expressions 7.20 and 7.21 above, for the case of different errors on each point (i.e. the weighted case).

- So what is special about the **Hessian** matrix? The term Hessian is actually used to define any square matrix that contains **second order partial derivatives**, and was first introduced by Ludwig Otto Hesse. For our particular case, the Hessian takes the form,

$$H_{jk} \equiv \frac{1}{2}\frac{\partial^2\chi^2}{\partial\alpha_j\partial\alpha_k} \tag{8.14}$$

where j and k denote the row and column in the case of the Hessian matrix respectively, and in the case of the single column parameter matrix α, they denote the elements in the column. We can check this for our example of the straight line fit. First taking H_{11} by differentiating Equation 8.5 wrt A, to get,

$$H_{11} \equiv \frac{1}{2}\frac{\partial^2\chi^2}{\partial A^2} = \sum 1/\sigma^2 \tag{8.15}$$

and H_{22} by differentiating Equation 8.6 wrt B, to get,

$$H_{22} \equiv \frac{1}{2}\frac{\partial^2\chi^2}{\partial B^2} = \sum x^2/\sigma^2 \tag{8.16}$$

and finally we can either take the derivative of 8.6 w.r.t. A, or the derivative of 8.5 w.r.t. B, to obtain the off-diagonal terms in the Hessian matrix,

$$H_{12} = H_{21} \equiv \frac{1}{2}\frac{\partial^2\chi^2}{\partial A\partial B} = \sum x/\sigma^2 \tag{8.17}$$

- The inverse of the Hessian matrix also has another feature: it is equivalent to the **variance covariance matrix**!

$$\mathrm{Cov}(\alpha_j, \alpha_k) = H^{-1}_{jk} \tag{8.18}$$

We see that the diagonal terms ($j = k$) are the standard variance terms and the off-diagonal terms hold the covariance of one parameter with another.

- Note the Hessian matrix represents the **curvature** of the χ^2 surface/manifold. In the case of **linear** functions, the Hessian matrix is **independent** of the fit parameters α. In the case of non-linear models, the Hessian matrix is a function of the fit parameters – i.e. the curvature of the χ^2 surface is dependent on the choice of fit parameter. In addition, the χ^2 surface is parabolic. You can see now that non-linear models can not be solved simply by inverting the Hessian. This is a tricky problem, and we'll come back to it in later.

- We mentioned above that if one can orthogonalise the normal equations then the fit parameters become independent of one another. We have also seen that for the standard case of the straight line fit $y = A + Bx$ that this is generally not the case, and that the parameters are dependent on one another, as we demonstrated in by analysing the χ^2 surface in Figure 8.2. From what we've just discussed, this should mean that the inverse of the Hessian will have off-diagonal terms, i.e. there exists covariance between the fit parameters A and B. We see from expression 8.12 that this is the case, and the value for the parameter dependence is given by

$$cov_{AB} = \frac{1}{\Delta} \sum \frac{x}{\sigma^2} \tag{8.19}$$

- So do we orthogonalise this fit? We mentioned above that simply replacing x with $x - \hat{x}$ would do this trick! Let's explore why. First, our χ^2 expression becomes,

$$\chi^2 = \sum \frac{(y - A - B(x - \hat{x}))^2}{\sigma^2} \tag{8.20}$$

then we can take the first derivative wrt to A,

$$\frac{\partial \chi^2}{\partial A} = -2 \sum \frac{(y - A - B(x - \hat{x}))}{\sigma^2} \tag{8.21}$$

and then by taking the derivative wrt to B, we get

$$\frac{\partial^2 \chi^2}{\partial A \partial B} = 2 \sum \frac{(x - \hat{x})}{\sigma^2}, \tag{8.22}$$

which give the value of the off-axis terms in the Hessian matrix. Interestingly we see that the numerator in the above expression for $\frac{\partial^2 \chi^2}{\partial A \partial B}$ is zero, since the sum of x is simply $\hat{x}$. So by replacing x with $\hat{x}$ we have effectively diagonalised the Hessian matrix. If the Hessian is diagonal, then so too is its inverse, and so the covariance terms are also zero – the fit parameters no longer depend on one another.

- It is also worth pointing out that the eigenvectors of the Hessian matrix define the principle axis of the χ^2 space. We show such a case for the χ^2 space that results from the straight line fit in Figure 8.6, where the principle axes of the ellipse are shown. Here we see that these principle axis are rotated w.r.t. to the original co-ordinate system. The processes of orthogonalisation effectively rotates the principle axis such that they align with the parameter axis. Remember, this is done by diagonalising the Hessian matrix as above.

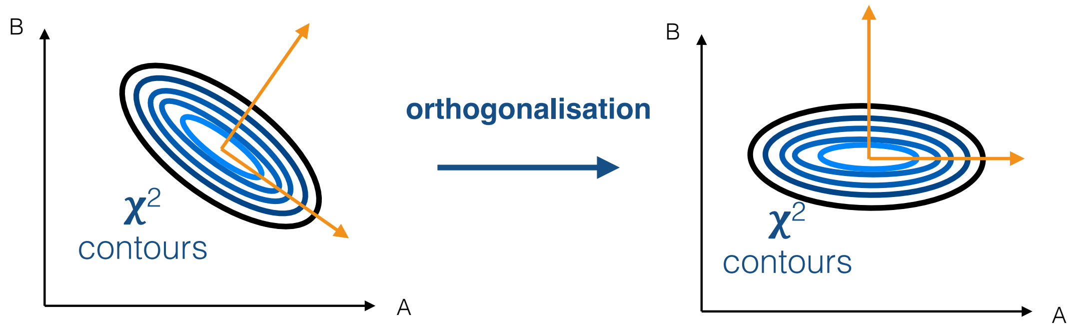

Figure 8.6:

8.4 Generalised linear regression

In the examples above, we have focused mainly on the simplified case of a straight line fit. Now we consider the more general form.

- The straight line fit can be thought of as a combination of two "patterns":

$$a_1 P_1(x) \tag{8.23}$$

and

$$a_2 P_2(x) \tag{8.24}$$

where the first pattern has coefficient $a_1 \equiv A$ and form $P_1(x) = 1$, and the second has coefficient $a_2 \equiv B$ and form $P_2(x) = x$. We now want to consider the more general case where,

$$y(x) = a_1 P_1(x) + a_2 P_2(x) + \cdots + a_M P_M(x) \tag{8.25}$$

$$= \sum_k^M a_k P_k(x) \tag{8.26}$$

- We can derive the normal equations from this as follows. First, we can define χ^2 to be,

$$\chi^2 = \sum_{i=1}^N \left[\frac{y_i - y(x_i)}{\sigma_i}\right]^2 \tag{8.27}$$

$$= \sum_{i=1}^N \frac{1}{\sigma_i^2} \left[y_i - \sum_k^M a_k P_k(x_i)\right]^2 . \tag{8.28}$$

We then get the M normal equations by taking the taking the derivative of the above expression for χ^2 wrt to each a_j, and setting the derivative to zero in the usual way,

$$0 = \frac{\partial \chi^2}{\partial a_j} = -2 \sum_i^N \frac{1}{\sigma_i} \left[y_i - \sum_k^M a_k P_k(x_i)\right] P_j(x_i). \tag{8.29}$$

Each term in the $k-$th normal equation is given by expanding out the j terms, with the usual sum over all the the N points (remember that in the last few examples above, we dropped those terms). We can then rearrange this to get the final version the generalised normal equations:

$$\sum_k^M \left[\sum_i^N \frac{P_k(x_i)P_j(x_i)}{\sigma_i^2} \right] a_k = \sum_i^N \frac{y_i P_j(x_i)}{\sigma_i^2}. \tag{8.30}$$

We see that each j will define a full normal equation, which has the individual terms denoted by k. This has the same form as,

$$\sum_k^M H_{jk} a_k = C_j(y) \tag{8.31}$$

i.e. j denotes the row of the Hessian, and the k denotes the column, such that,

$$H_{jk} = \sum_{i=1}^N \frac{P_j(x_i)P_k(x_i)}{\sigma_i^2} = \frac{1}{2} \frac{\partial^2 \chi^2}{\partial a_j \partial a_k} \tag{8.32}$$

- This can be easily packaged up into a computer code, and an off-the-shelf matrix inversion routine can be used to solve for a_j. One can also use standard routines to diagonalise the Hessian matrix, to reduce the interdependency in the fit parameters.
- Note that the errors in the fit parameters obtained from this process are only correct if the errors in the data are strictly **Gaussian / normal**. We will discuss in later lectures how to deal with situations where this is not the case.

8.5 Scaling functions to fit data

- The process of χ^2-minimisation is also useful for scaling a function to fit a data set. More generally, we can use χ^2-minimisation to scale a pattern P to fit the data. This pattern could be an analytic function, or simply just another data set (e.g. in the case of spectroscopy, a template spectrum).
- Consider a pattern $y = P(x)$ that we wish to scale to a data set (x_i, y_i), which contains N observations. If we have hetroscedastic, but normally distributed errors, then we can write,

$$\chi^2 = \sum_i^N \left(\frac{y_i - AP(x_i)}{\sigma_i} \right)^2 \tag{8.33}$$

Note here that although P is function, we know its shape. What we don't know is how to scale it, and that is controlled by the parameter A: this is what we want to find. We therefore perform the standard χ^2 minimisation w.r.t. A:

$$0 = \frac{\partial \chi^2}{\partial A} = -2 \sum_i^N \frac{[y_i - AP(x_i)]\, P(x_i)}{\sigma_i^2} \tag{8.34}$$

which gives,

$$\sum_i^N \frac{y_i\,P(x_i)}{\sigma_i^2} = \sum_i^N \frac{A\,[P(x_i)]^2}{\sigma_i^2} \tag{8.35}$$

$$A = \frac{\sum_i^N y_i\,P(x_i)/\sigma_i^2}{\sum_i^N [P(x_i)]^2/\sigma_i^2} \tag{8.36}$$

- We can also get an error on the scale-factor A by looking at the second derivative of χ^2 (here our Hessian matrix is 1×1!). This is given by,

$$\frac{\partial^2\chi^2}{\partial A^2} = +2\sum_i^N \frac{[P(x_i)]^2}{\sigma_i^2} \tag{8.37}$$

and so the variance is given by

$$\sigma_A^2 = \frac{2}{\frac{\partial^2\chi^2}{\partial A^2}} = \frac{1}{\frac{\sum_i^N [P(x_i)]^2}{\sigma_i^2}} \tag{8.38}$$

8.6 Non Linear Fitting

- In non-linear models, the Hessian matrix is no longer independent of the fit parameters α, and as such, there is generally no analytic solution for the model, and so we lose the elegance that we had in the linear case, in which the normal equations can be simply solved to give the 'best' fit model parameters.

- We also mentioned that the Hessian matrix represents the curvature of the χ^2-space. Since the Hessian matrix now depends on the fit parameters, the contours in χ^2-space are now banana shaped. We show an example of this in Figure 8.7. The χ^2-space can also have multiple peaks, and so more care needs to be taken in using the χ^2-space to make decisions about α.

- For easy problems, with just 2, or perhaps 3 variables, we can simply grid up the parameter space, calculate χ^2 in the usual manner, and inspect the χ^2 surface to find the best-fit parameters. But for functions of more variables, this starts to become unwieldy, and we need to look to other, more sophisticated methods.

- However, all is not lost, and there are several approaches that one can take to solve for α.

  The most common method is simply to **linearise the non-linear model**. Basically we are saying that over a very small region close to the best fit values of α, the model can be thought of as approximately linear. Take for example, a classic problem that arises

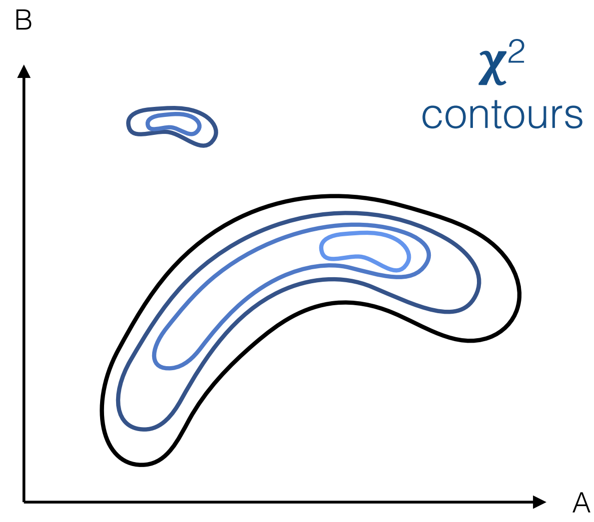

Figure 8.7:

in spectroscopy, in which one is trying to fit a gaussian line, to a spectrum in which there is a background, i.e.

$$y = Ae^{-\eta^2/2} + B \tag{8.39}$$

where $\eta = (x - x_0)/\sigma$. So now we have 4 fit parameters: A, B, x_0 and σ. Don't mistake this σ for the error in the points! It represents the width of the line. To linearise this function, we simply take the first-order terms from a Taylor expansion around a guess at the parameter values, which we will denote $\hat{A}$, $\hat{B}$, $\hat{x}_0$, and $\hat{\sigma}$. Our linearised function now becomes,

$$y = \hat{y} + (A - \hat{A})\frac{\partial y}{\partial A} + (B - \hat{B})\frac{\partial y}{\partial B} + (x_0 - \hat{x}_0)\frac{\partial y}{\partial x_0} + (\sigma - \hat{\sigma})\frac{\partial y}{\partial \sigma} \tag{8.40}$$

where $\hat{y} = y(\hat{A}, \hat{B}, \hat{x}_0, \hat{\sigma})$ and the derivatives are

$$\frac{\partial y}{\partial A} = e^{-\eta^2/2} \tag{8.41}$$

$$\frac{\partial y}{\partial B} = 1 \tag{8.42}$$

$$\frac{\partial y}{\partial x_0} = \frac{A\eta}{\sigma} e^{-\eta^2/2} \tag{8.43}$$

$$\frac{\partial y}{\partial \sigma} = \frac{A\eta^2}{\sigma} e^{-\eta^2/2} \tag{8.44}$$

and and are obviously evaluated at the point $\hat{y}$. Since the derivative terms are constant at $\hat{y}$, and $\hat{A}$, $\hat{B}$, $\hat{x}_0$, and $\hat{\sigma}$ are known (i.e. guessed!), we now have model that is linear in the fit parameters. We can now proceed as usual to get values of A, B, x_0 and σ from the normal equations. Once the new values are found, we can repeat the above process, using the new point as the 'guess' point to define $\hat{y}$. This iteration is continued until the values converge. For functions with low numbers of variable, such as the Gaussian + background example given here, this technique is fairly powerful. Also, for many cases, such as in line fitting, it is often easy to make a good guess at the fit parameters to ensure the iteration is fast. However for strongly peaked functions, this type of fit can be dangerous, since the iteration may not converge. Generally speaking it is a good idea to view the χ^2 surfaces in advance!

- The last method that we will discuss for exploring the non-linear χ^2 surface should be familiar to you: **MCMC**! This has a number of advantages:

  1. Some of the heartache of tuning the width of the proposal distribution is not necessary here. Remember that the variance-covariance matrix is the inverse of the Hessian matrix, which in turn was the curl of the χ^2 surface. We can therefore use the terms σ_A, σ_B, etc. that arise from the variance-covariance matrix to decide on the width of the proposal distribution in each direction (i.e. along each parameter axis). In this way, the jumping length always adapts to the local conditions.
  2. Another advantage of the MCMC is that MCMC can escape from local minima.
  3. The MCMC output will be a list of positions in parameter space, and their relative probability, which should provide a good coverage of the parameter space. If one is using explicit maximum likelihood, in place of χ^2 minimisation, then one can also add any prior information to create a full posterior distribution. Clearly in this case, one would need to use a different estimate of the width of the proposal distributions.

  Be careful when using MCMC in the χ^2 minimisation case! Remember that our previous MCMC discussion before involved unconditionally accepting those proposed steps that have a greater value of f than the current location – now the situation is reversed, since we want to find the **lowest** value of χ^2! In the case of maximum likelihood (either with or without prior), then we obviously use the standard MCMC formalism.

8.7 Bootstrapping – getting something for nothing!

- In the above discussion, we were careful not to discuss the errors in the fit parameters in the non-linear case. The reason for this is that the variance-covariance matrix is now an approximation to the underlying errors. And remember that the analytic expressions for the errors are **only** valid in the case where the errors are strictly normal – even in the case of the simple log-ed form of a power-law fit, this is no longer true, and the analytic terms need to be viewed with considerable caution.

- So how to proceed? Thankfully we can use a process called **bootstrapping** to estimate the errors. The bootstrapping process is extremely simple and involves the following steps:

  1. For a data set with N data points, $X = x_1, x_2, \ldots, x_N$,we simply take N random draws from our data to create a new, fake, data set. Since there will probably be several points that appear more than once, and therefore also points which are missing entirely, this is termed **resampling with replacement**. For example, if we had 8 points, our first resample may look like,

  $$X_1^* = \{x_3, x_6, x_8, x_3, x_4, x_5, x_1, x_1\} \tag{8.45}$$

     where the notation X^* denotes a bootstrap resampling of the data set X. In this example we see that x_3 and x_1 appear twice, but there are no instances of x_2 or x_7 in the bootstrapped data set.
  2. With the new data set, we can now take the mean of our sample in the usual way to get a new estimate of the mean, $\hat{X}_1^*$.
  3. We repeat steps 1 and 2 many times, to build up a distribution of estimates of the mean from the data. From this we can compute an expectation value for the mean, and it's variance.

- So how many times to we need to resample the data? There is a lot of literature on this, but typically the answer is, as many times as you reasonably can! Clearly even a small number of resamplings can provide information about the mean might vary that you didn't have before, That said, you don't want a few unlucky resampling events to bias your beliefs and so it is generally recommended to resample at least 1000 times. On modern computers this is normally straightforward.

- In non-linear function fitting, we can use the bootstrap technique to fake new datasets. We then fit each data set in turn, to get an estimate for how the fit parameters might vary. As you can imagine, this can be a slow process, since each fit may take some time to converge, but for non-linear functions it provides a relatively convenient way, and in some cases, the only way, to estimate the errors in the fit. Given that the coding is simple, the technique has been widely adopted. It is also used extensively in linear regression for cases where there are poor error constraints on the data points.

- With bootstrapping it might seem that you are getting something for nothing! Indeed, it took the statistics community around 20 years to work out the formal mathematics that justifies its use.